

Ahmad Mustafa Anis

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PROFESSIONAL SUMMARY

AI/ML Research Engineer with 4+ years of industry experience building and deploying deep learning systems across computer vision, video understanding, and natural language processing. Published at CVPR, ICLR, and NeurIPS; currently conducting research in Self-Supervised Learning at Carnegie Mellon University's TarrLab. Experienced across the full research-to-production pipeline, from model architecture and fine-tuning to real-time inference optimization and multimodal system design.

EDUCATION

Carnegie Mellon University

Master of Science in Artificial Intelligence Engineering.

Pittsburgh, PA

Dec 2027

International Islamic University

Bachelor of Science in Computer Science.

Islamabad, Pakistan

Sep 2022

EXPERIENCE

Research Assistant

Jun 2026 – Present

TarrLab, Carnegie Mellon University

- Investigate progressive learning strategies to improve Self-Supervised Learning, supervised by Professor Michael Tarr and Professor Deva Ramanan.

Deep Learning Engineer

Mar 2024 – Aug 2026

Roll AI

- Optimized real-time video bokeh effects by implementing Depth Estimation and Matting algorithms, achieving 2x performance improvement through parallel processing and batch optimization for live streaming applications.
- Developed and deployed an Active Speaker Detection system for multi-camera live events, automating speaker-camera associations and reducing event video editing time from 48 hours to 1 hour (98% improvement).
- Built a Video Understanding pipeline for automated chapter generation and captioning, enabling intelligent b-roll suggestions by analyzing semantic content and scene transitions across long-form video.
- Developed a multi-view video generation pipeline that synthesizes novel viewpoints from temporal endpoints, enabling enhanced scene reconstruction for live event production.
- Architected a multimodal LLM Agent as a centralized AI video editing system, automating social media clip generation, multi-take video orientation and cropping, transcript correction and enhancement, and dynamic VFX and motion graphics creation to streamline post-production workflows.

AI Research Fellow

Sep 2024 – Jun 2025

Fatima Institute of Global AI Research

- Researched Retrieval-Augmented Test Time Inference for Medical Vision Language Models, advised by Dr. Wei Peng from Stanford University.

AI Fellow

Dec 2023 – Feb 2024

PI School of AI

- Selected as 1 of 10 fellows from 300 global applicants for a fellowship valued at 12,500 EUR; extracted key information from long-context interview transcripts using LLMs (MistralLite, Mixtral8x7B) and built RAG pipelines using DSPy.

Machine Learning Engineer

Apr 2022 – Mar 2024

Red Buffer

- Built a fact verification and identification pipeline with multi-threading for a 200% speed increase; fine-tuned LLaMAv2, Dolly-3B, GPT-Neo, and BART for negation generation to detect false news.
- Reduced OCR latency from 0.6 FPS to 16 FPS on CPU for streaming devices to blur sensitive information.
- Boosted logo recognition accuracy from 30% to 85% during cricket matches using CLIP by OpenAI.
- Developed a drag-and-drop RAG system compatible with any LLM, vector DB, and multi-hop search.

Deep Learning and Computer Vision Intern

Jul 2021 – Dec 2021

WortelAI

- Developed a Safety Clothes Detection system for factory workers using YOLOv5.
- Trained a Celebrity Detection and Classification system using 3DDFAv2, ResNet50, and a 172GB dataset.
- Built a real-time video search engine using CLIP by OpenAI, Qdrant for vector DB, and ElasticSearch.
- Classified multi-label text using BERT and PyTorch across 1,500+ labels.

PROJECTS

DistAYA

- Contributed to the AYA Expedition project focused on creating small multilingual LLMs distilled from large multilingual LLMs, published on Hugging Face.

PUBLICATIONS

- A. M. Anis, Hasnain Ali, M. Saquib Sarfraz. On the Limitations of Vision-Language Models in Understanding Image Transforms. CVPR 2025 Workshop: Computer Vision in the Wild.
- S. Longpre, N. Singh, ... A. M. Anis (11th Author), et al. Bridging the Data Provenance Gap Across Text, Speech, and Video. ICLR 2025.
- S. Longpre, R. Mahari, ... A. M. Anis (16th Author), et al. Consent in Crisis: The Rapid Decline of the AI Data Commons. NeurIPS 2024.

OPEN SOURCE CONTRIBUTIONS

- Pandas (pandas-dev/pandas): Contributed bug fixes and improvements to the core data manipulation library.
- LlamaIndex (run-llama/llama_index): Contributed to the open-source LLM application framework.
- Google Research BigVision (google-research/big_vision): Contributed to the large-scale vision research codebase.
- OpenAI Shap-E (openai/shap-e): Contributed to the text-to-3D generation model codebase.

INTERVIEWS, PODCASTS, AND TECHNICAL WRITING

- Interview with AtomCamp: Provided advice for beginners in Machine Learning.
- Podcast, Adventures in Machine Learning: Discussed optimizing and improving OCR systems.
- Podcast, Adventures in Machine Learning: Discussed deploying Deep Learning models with FastAPI.
- Technical writing published on Medium, KDNuggets, and CNVRG.

ACTIVITIES & LEADERSHIP

Data Provenance Initiative, MIT Media Lab <i>Research Contributor</i>	Dec 2023 - Present
• Audit LLM training data to assess web data access shifts, cultural representation, and transparency.	
Cohere Labs Community <i>Volunteer Researcher and Regional Community Lead</i>	Nov 2022 - Present
• Organized 50+ guest speaker sessions with Asian researchers; initiated NYU Deep Learning study group (Prof. Yann LeCun); served as Urdu Language Lead for AYA and contributed 3 Urdu datasets.	
MESA Program at Foothill College <i>Mentor</i>	Jan 2024 - Present
• Provide mentorship to underrepresented students in STEM.	
Bytewise Limited <i>Fellowship Lead (NLP)</i>	Jun 2024 - Aug 2024
• Mentored 20 students and supervised the NLP curriculum.	
Amazon Web Services <i>AWS Community Builder</i>	Aug 2022 - Apr 2023
• Organized community events to raise awareness about AWS services.	

SKILLS

Libraries and Frameworks: PyTorch, PyTorch Lightning, Distributed Training, TensorFlow, OpenCV, FastAPI, Flask
Programming Languages: Python, C++, Rust, Java, JavaScript
Databases: MySQL, PostgreSQL
Tools: Git, GitHub, GitLab, Docker, Jupyter Notebook, Anaconda, Visual Studio Code

CERTIFICATIONS

- Microsoft Technology Associate (MTA) 98-381: Introduction to Programming Using Python
- Machine Learning - Coursera
- Deep Learning Specialization - Coursera
- Getting Started with TensorFlow 2 - Coursera
- NYU Deep Learning by Yann LeCun
- Mathematics for Machine Learning - Imperial College London

AWARDS & HONORS

- 2,600+ points on StackOverflow, impacting 300,000+ people
- 5 kyu, top 36% worldwide on Codewars
- Platinum badge on KDNuggets